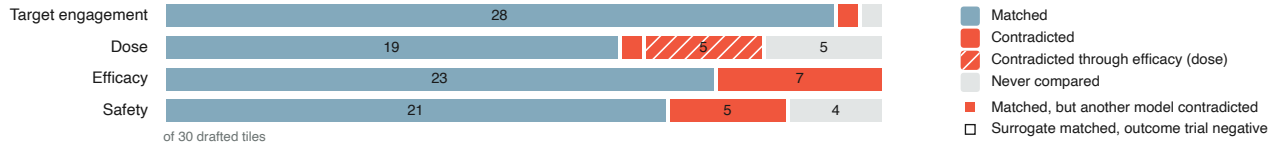
a



Panel b: every drafted tile; diseases by global DALYs in 2021 (WHO Global Health Estimates), then those without a WHO cause by area.

b

		Target engagement	Dose	Efficacy	Safety
#1	**type 2 diabetes mellitus** GLP1R agonists (12)	**2011** Cynomolgus monkeys and rat islets (dulaglutide)	**2019** Beagle dog oral bioavailability of semaglutide/SNAC tablets	**2023** GLP1R Gly168Ser genotype, GLP-1 infusion clamp in healthy volunteers	**2016** GLP1R Ala316Thr carriers (human genetics)
#1	**type 2 diabetes mellitus** GCK activators (9)	**2011** Rodent models of type 2 diabetes	**2012** Glucose-insulin-glucagon model (LY2599506) Dose reached; effect faded	**2019** People with GCK-activating or GKRP loss-of-function variants AZD1656 missed 4-month HbA1c	**2024** Rodent models of type 2 diabetes GCK proxy misses GKA lipid rise
#3	**HIV** HIV integrase (INSTI) inhibitors (3)	**2011** HIV-1 replication in cell culture (MT-4/PBMC)	**2013** Protein-adjusted cell-culture potency as plasma trough target	**2012** In-vitro susceptibility of INSTI-resistant HIV-1 (site-directed mutant clones, fold-change)	**2026** Rat and rabbit embryo-fetal development studies Mouse NTD; human signal gone
#4	**NSCLC** EGFR inhibitors (18)	**2017** T790M/sensitising-mutant NSCLC cells and mice	**2019** PC9 brain-metastasis mice; monkey osimertinib PET	**2023** Patient-derived lung cancer organoids EXCLAIM-2 fails primary PFS	**2015** hERG, rat and dog tox, dog telemetry QT prolongation the package cleared
#4	**NSCLC** anti-IGF1R antibodies (4)	**2011** Tumour cells and xenografts given h7C10	**2005** A549 NSCLC xenografts in nude mice	**2011** A549 NSCLC xenografts in nude mice R1507 + erlotinib: no PFS gain	**2021** Cynomolgus repeat-dose tox (R1507) Monkeys missed hyperglycemia
#5	**migraine disorder** CGRP receptor antagonists (4)	**2014** Rhesus capsaicin-induced dermal vasodilation (CIDV)	**2014** Rhesus CIDV PK/PD target Predicted 20 mg; needed 200 mg	**2013** Primate CGRP-induced facial blood flow (rimegepant / BMS-927711)	**2026** Mouse MCAO stroke, gepant pretreatment Animal tox missed MK-3207 liver
#6	**Alzheimer disease** BACE1 inhibitors (4)	**2015** PDAPP mice and cannulated beagle dogs (brain/CSF Aβ)	**2019** Dog CSF Aβ PK/PD scaled to humans Target reached, no benefit	**2019** A673T variant carriers Genetics said protect; drug worsened	**2019** Mice chronically dosed with BACE1 inhibitors Liver injury missed by tox
#6	**Alzheimer disease** anti-APP antibodies (3)	**2021** tg-ArcSwe mice, mAb158 Mice: no plaque fall; patients: PET fell	**2014** hAPP/PS1 mice, crenezumab 15 mg/kg failed; higher doses failed too	**2014** hAPP/PS1 mice, crenezumab Crenezumab missed both primaries	**2017** Aged APP mice given 2H6 (ponezumab parent) Excess microbleeds missed
#7	**asthma** PTGDR2 antagonists (9)	**2016** Human eosinophils and Th2 cells in vitro	**2020** Eosinophil DP2 potency Target reached; exacerbations unchanged	**2014** Allergen challenge Challenge said yes; FEV1 trials missed	**2012** Off-target selectivity panels; animal toxicology unpublished
#8	**breast cancer** CDK4/6 inhibitors (6)	**2016** Human tumour xenografts in mice (phospho-Rb Ser780)	**2017** Abemaciclib xenograft PK/PD (pRb to growth) Brain exposure OK, no response	**2014** Human breast cancer cell-line panel (47 lines) Brain-metastasis trial missed	**2017** Dog cardiovascular telemetry (QTc) with hERG assays Missed liver injury and clots
#13	**schizophrenia** PDE10A inhibitors (5)	**2019** Rat and monkey ketamine-challenge phMRI/EEG reversal	**2019** Rodent occupancy dosing Predicted-adequate exposure, no effect	**2017** Ketamine challenge Challenge passed; phase 2 PANSS missed	**2017** Rat catalepsy, prolactin and glucose vs D2 antagonists EPS in patients at clinical dose
#15	**hepatitis C** HCV NS3/4A protease inhibitors (13)	**2013** Chronically HCV-infected chimpanzee (genotype 1a/1b)	**2009** Replicon EC99 vs rat plasma and liver exposure	**2016** Human hepatocyte chimeric mouse (uPA/SCID) infected with patient HCV genotype 1b	**2016** Rodent and dog repeat-dose toxicology
#17	**acute myeloid leukaemia** FLT3 inhibitors (4)	**2013** FLT3-ITD cell lines and MV4-11 xenografts (phospho-FLT3)	**2013** FLT3-ITD cell lines and MV4-11 xenografts (phospho-FLT3)	**2012** Ba/F3 FLT3-ITD resistance screen Combination: no survival gain	**2018** Rat and dog toxicology and safety pharmacology PRES and QT missed by tox species
#19	**MASH** NR1H4 agonists (4)	**2019** Rodent FXR target-gene PD (tropifexor)	**2019** STAM and AMLN-diet NASH mice (tropifexor)	**2021** STAM and AMLN-diet NASH mice (tropifexor) Outcome trial negative	**2024** High-fat-diet mouse ALT/cholesterol screen Cilofexor: no lipid change
#20	**prostate cancer** AR antagonists (5)	**2013** LNCaP/AR castrate xenografts	**2013** LNCaP/AR castrate xenografts	**2016** LNCaP/AR castrate xenografts	**2019** Mouse brain:plasma distribution with rat and dog CNS toxicology (darolutamide)
#22	**ovarian cancer** PARP1/2 inhibitors (6)	**2009** Human tumour xenografts, needle-biopsy PAR ELISA (veliparib)	**2021** NOVA exposure–safety model (niraparib)	**2016** BRCA-deficient cancer cells and xenografts (niraparib)	**2017** Dog cardiovascular safety pharmacology (niraparib)
#23	**rheumatoid arthritis** BTK inhibitors (8)	**2013** BTK occupancy probe in CIA mice	**2022** CIA occupancy-efficacy PK/PD Target reached, no effect	**2019** Collagen-induced arthritis (rodent) CIA-active drug failed Phase 2	**2021** Human platelet aggregometry and humanized-mouse thrombus formation (acalabrutinib vs ibrutinib)
#24	**multiple myeloma** TNFRSF17-directed depleters (8)	**2016** BCMA+ myeloma xenografts in NSG mice, CAR-T	**2014** BCMA+ myeloma xenografts in mice, ADC	**2019** bb2121 CAR T cytotoxicity against cell lines of quantified BCMA density (in vitro)	**2020** Rat and rabbit repeat-dose toxicology (belantamab mafodotin)
#27	**open-angle glaucoma** ROCK inhibitors (3)	**2018** Living mouse eye, OCT of conventional outflow tissue, topical netarsudil	**2015** Normotensive monkey eye, topical netarsudil 0.04%	**2015** Normotensive monkey eye, topical netarsudil 0.04% Genotype did not predict response	**2017** Monkey 9-month topical ocular toxicology
Autoimmune	**psoriasis** RORC inverse agonists (4)	**2021** Human whole-blood IL-17 secretion assay No IL-17 change in plaques	**2021** Human whole-blood IL-17 secretion assay	**2022** IL-23 minicircle mouse, oral BI 730357	**2015** Nonclinical toxicology species (not public)
Autoimmune	**psoriasis** anti-IL17A antibodies (4)	**2016** Mice injected with human IL-17A, plasma KC (CXCL1) readout	**2016** Cynomolgus repeat-dose toxicology	**2023** Mice injected with human IL-17A, plasma KC (CXCL1) readout	**2016** People with inborn errors of IL-17 immunity
Autoimmune	**ulcerative colitis** S1PR1 agonists (3)	**2016** Rodent blood lymphocyte counts (ozanimod)	**2023** Rat and dog repeat-dose toxicology (etrasimod)	**2016** TNBS and CD4+CD45RBhi transfer colitis mice (ozanimod)	**2023** hERG current assay in vitro (etrasimod) Ozanimod still slows the heart
GI	**GERD** ATP4A blockers (3)	**2019** Stimulated gastric acid secretion in dogs	**2017** Stimulated gastric acid secretion in dogs	**2019** Rat reflux oesophagitis vs esomeprazole	**2025** Rat repeat dosing: gastrin, emptying, microbiota
Hematology	**hemophilia B** F9 replacement / gene therapy (7)	**2017** Hemophilia B dogs, AAV8 canine FIX-Padua FIX expression lost in 7 of 8	**2017** Hemophilia B dogs, AAV8 canine FIX-Padua	**2024** Juvenile hemophilia B dogs, canine surrogate vector	**2024** Cynomolgus GLP tox, fidanacogene predecessor
Hematology	**PNH** anti-C5 antibodies (4)	**2022** Hemolysis assay with C5 R885H variant	**2020** Cynomolgus PK/PD (crovalimab)	**2018** Complement hemolysis assay in human serum	**2020** Drug–target–drug complex assay, human C5
Met.	**hyperlipidemia** CETP inhibitors (3)	**2014** Human CETP/apoA-I transgenic mouse	**2014** CETP/apoA-I mouse; rat blood pressure Exposure far below prediction	**2011** Human CETP loss-of-function genotypes Outcome trial negative	**2014** Rat blood pressure + human adrenal H295R cells
Psy.	**insomnia** HCRTR1/2 antagonists (6)	**2013** Rat, dog and rhesus sleep EEG/EMG (suvorexant)	**2022** PBPK-PD model of brain OX2 receptor occupancy (daridorexant, 65% occupancy threshold)	**2012** Rat, dog and rhesus sleep EEG/EMG (suvorexant)	**2019** Mouse emotion-induced cataplexy (lemborexant) Animals no abuse; humans liked it
Resp.	**cystic fibrosis** CFTR correctors (10)	**2024** FRT cells expressing CFTR variants Third corrector added nothing	**2015** Rat and dog toxicology (lumacaftor)	**2024** FRT cells expressing CFTR variants	**2025** Rat and dog tox (vanzacaftor)
Uro.	**interstitial cystitis** anti-NGF antibodies (3)	**1997** Rodent chemical cystitis	**1997** Rodent chemical cystitis	**2022** ASP6294 vs anti-NGF IgG in arthritic rodents ASP6294 missed IC pain endpoint	**2012** Primate and rodent joint tox Joint harm missed; IC trials put on hold
Other	**atopic eczema** PDE4 inhibitors (5)	**2019** Stimulated human blood cells (PBMC cytokine release)	**2016** Minipig dermal toxicity (3- and 9-month)	**2023** Mouse chronic allergic contact dermatitis (difamilast) Explant engaged; primary failed	**2016** Rabbit primary skin irritation